CSE 332: Analog and Digital Circuits — Winter 2026

Final Project: Non-Invasive Wearable Biosensor for Rodent Research

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1. PCB Design Description

1.a. Overview

This project presents the design of a two-layer printed circuit board (PCB) for a non-invasive wearable biosensor intended for laboratory mice. The system continuously measures heart rate (HR), heart rate variability (HRV), respiration, body temperature, activity, and sleep without requiring surgery or a tethered power cable. Data is transmitted wirelessly in real time via Bluetooth Low Energy (BLE) to a gateway and cloud infrastructure.

The main microcontroller is the nRF52832-CIAA-R7 in a WLCSP package (3mm x 3.2mm), which integrates the ARM Cortex-M4 processor and the BLE radio into a single chip, minimizing board area. The biological sensors (MAX30101, TMP117, LSM6DSL) are placed on the bottom copper layer (B.Cu) to maintain direct contact with the rodent's skin, while the power management components and microcontroller are placed on the top layer (F.Cu).

1.b. Major Circuit Components and Selection Rationale

All components were selected to prioritize small form factor, low power consumption, and simple digital communication interfaces:

- nRF52832-CIAA-R7 (Microcontroller + BLE Radio): Chosen for integrating the processor and BLE radio in a single WLCSP chip (3mm x 3.2mm). The ESP32 was considered but rejected due to its larger size and higher power consumption. The WLCSP package was selected over the QFN variant (6mm x 6mm) to minimize board area.
- MAX30101EFD+T (Heart Rate/HRV/Respiration Sensor): Integrates LEDs, photodetectors, and low-noise analog front-end in a single package, measuring HR, HRV, and respiration via one I2C interface. The MAX30102 was rejected for lacking the green LED channel needed for respiration measurement.
- TMP117AIDRVR (Temperature Sensor): Provides ±0.1°C accuracy without calibration and communicates over I2C. Preferred over the LM75 for superior accuracy and smaller DRV package. Placed on B.Cu for direct skin contact.
- LSM6DSL (Accelerometer/Gyroscope): A 6-axis IMU enabling activity, sleep, and 3-axis motion detection. Supports both I2C and SPI and fits in a compact 3mm QFN package.
- BQ25100HYFPT (Battery Charger IC): Charges LiPo cells at up to 250mA with only 75nA battery leakage current in standby — critical for maximizing battery life during idle periods. The 'H' variant was selected for its JEITA profile optimized for warm body temperatures.
- TPS63031DSKR (Buck-Boost Regulator): Maintains a stable 3.3V output as the LiPo battery discharges from 4.2V down to 2.8V, achieving up to 96% conversion efficiency.

1.c. How the Design Meets Project Requirements

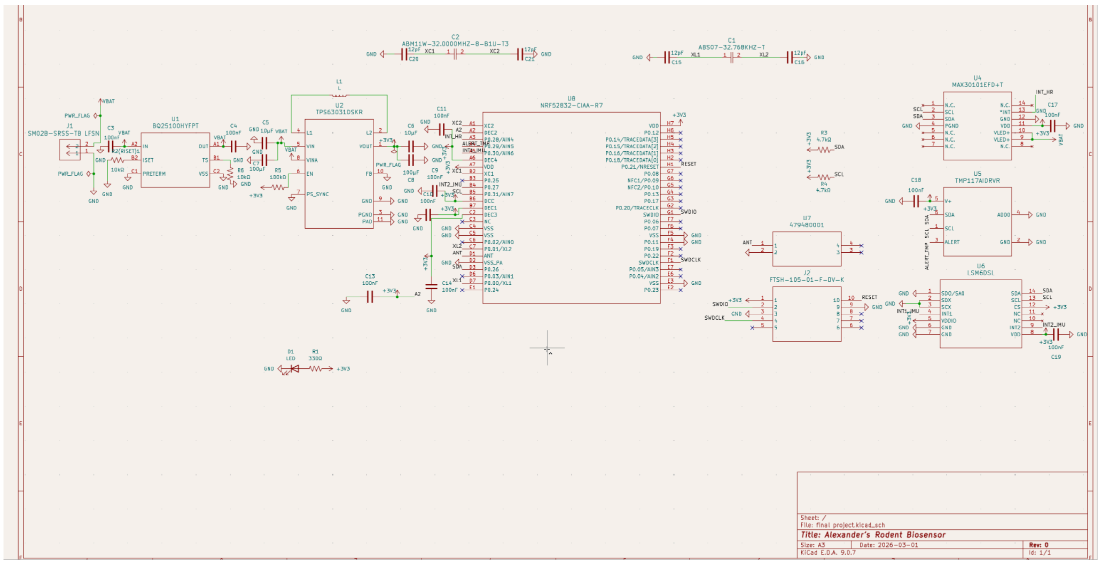
- Continuous non-invasive monitoring: Sensors on B.Cu are in direct skin contact with the rodent without surgery, enabling continuous data collection.
- Wireless real-time data transmission: The nRF52832 transmits sensor data via BLE to a gateway and cloud infrastructure in real time.
- No tethered power: The 80mAh LiPo battery provides autonomous power, rechargeable via the BQ25100 charger.
- Continuous temperature measurement: The TMP117 continuously measures body temperature with ±0.1°C precision.
- 3-axis specific force (accelerometer): The LSM6DSL provides 3-axis acceleration for activity and sleep detection.

- Board size: The PCB measures 27mm x 18mm. The ideal target of 20mm x 5.5mm is not achievable with commercial off-the-shelf components — the nRF52832 WLCSP alone is 3mm x 3.2mm, but the sum of all components exceeds the width target. A future iteration using custom ASICs could approach the target dimensions. This tradeoff is discussed further in Section 2.
- Weight: PCB + components approximately 1.5g, 80mAh battery approximately 1.55g, total approximately 3.05g. The 2.5g ideal target would require a custom-fabricated battery or a custom ASIC redesign.

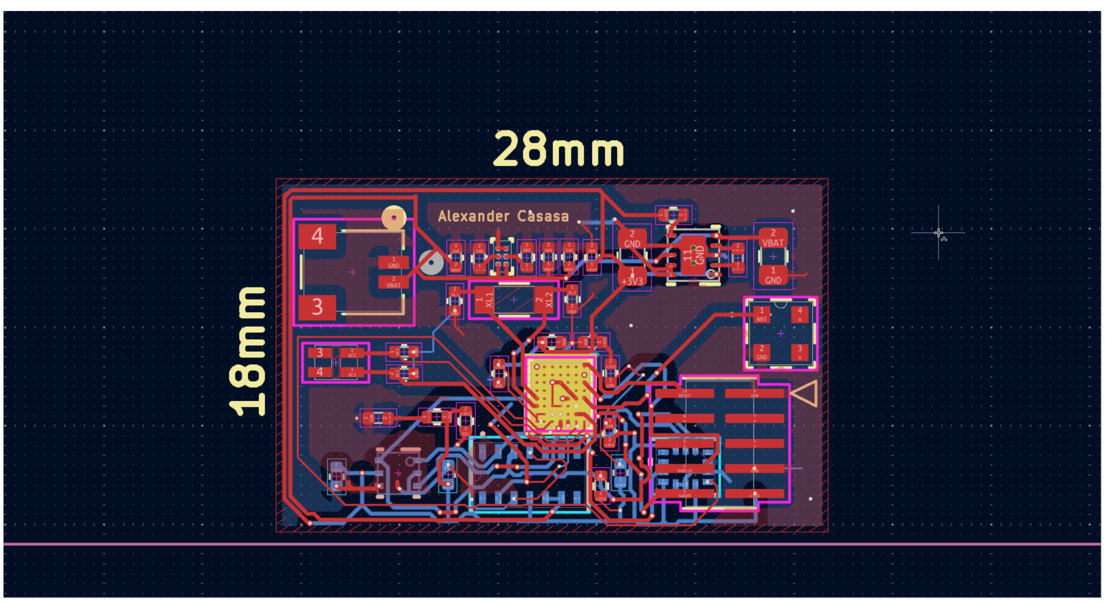
1.d. Figures

The following figures illustrate the complete schematic, PCB layout with board dimensions, and 3D view of the assembled board.

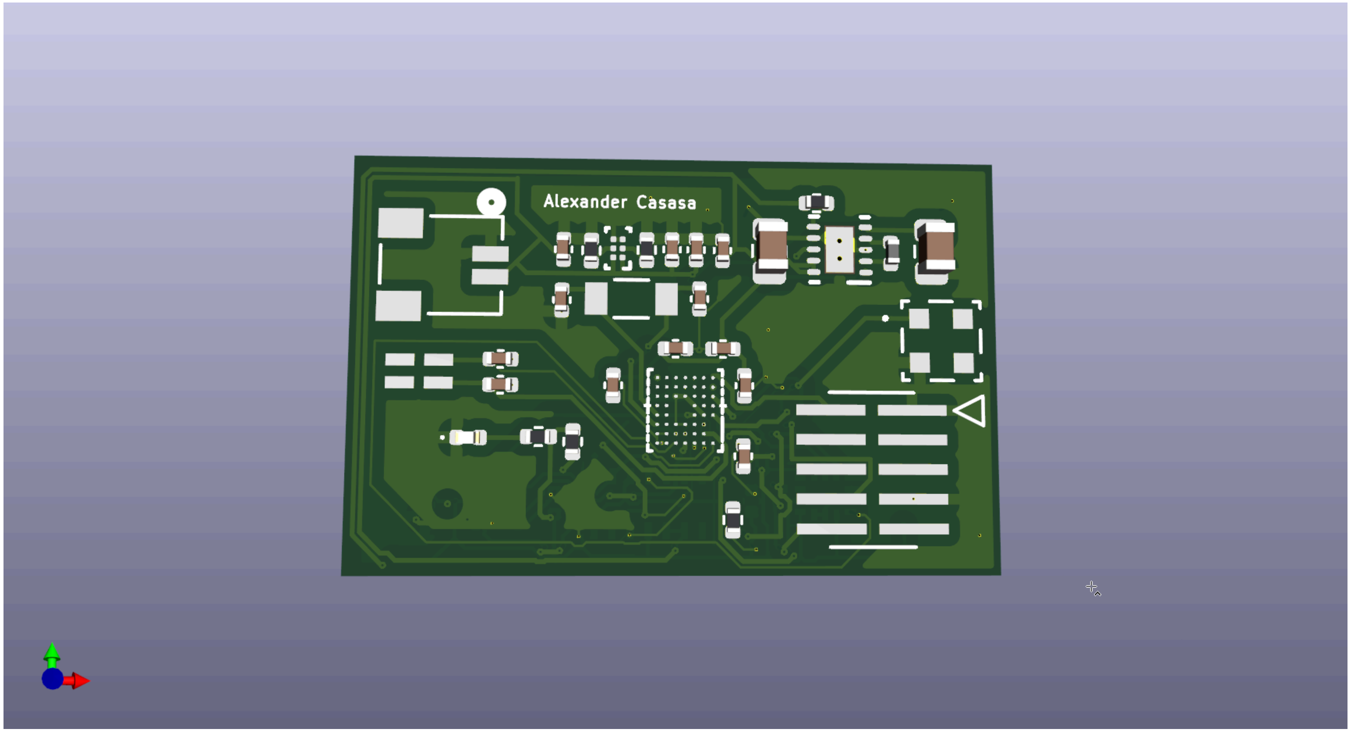
[Figure 1: Complete system schematic]



[Figure 2: PCB layout with dimensions (27mm x 18mm)]



[Figure 3: 3D view of assembled PCB]



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1.e. Bill of Materials

Ref.	Part Number	Component Type	Qty.	Notes
U1	BQ25100HYFPT	Battery Charger IC	1	Low-current LiPo charger, 75nA battery leakage, JEITA profile for warm environments
U2	TPS63031DSKR	Buck-Boost Regulator	1	Fixed 3.3V output, up to 96% efficiency, 1.8V-5.5V input range
U4	MAX30101EFD+T	Heart Rate/SpO2 Sensor	1	Integrated LEDs, photodetectors, low-noise AFE, I2C interface
U5	TMP117AIDRVR	Temperature Sensor	1	±0.1°C accuracy, no calibration needed, I2C interface
U6	LSM6DSL	Accelerometer/Gyroscope	1	6-axis IMU, activity/sleep detection, I2C/SPI interface
U7	479480001	2.4GHz Chip Antenna	1	SMD chip antenna for BLE, 3.2mm x 1.6mm footprint
U8	NRF52832-CIAA-R7	Microcontroller + BLE Radio	1	ARM Cortex-M4, integrated BLE, WLCSP package (3mm x 3.2mm)
J1	SM02B-SRSS-TB	JST Battery Connector	1	2-pin JST SH connector for LiPo battery
J2	FTSH-105-01-F-DV-K	SWD Debug Header	1	10-pin 1.27mm pitch ARM SWD programming header
C1	ABS07-32.768KHZ-T	32.768kHz Crystal	1	RTC crystal for nRF52832 low-power clock

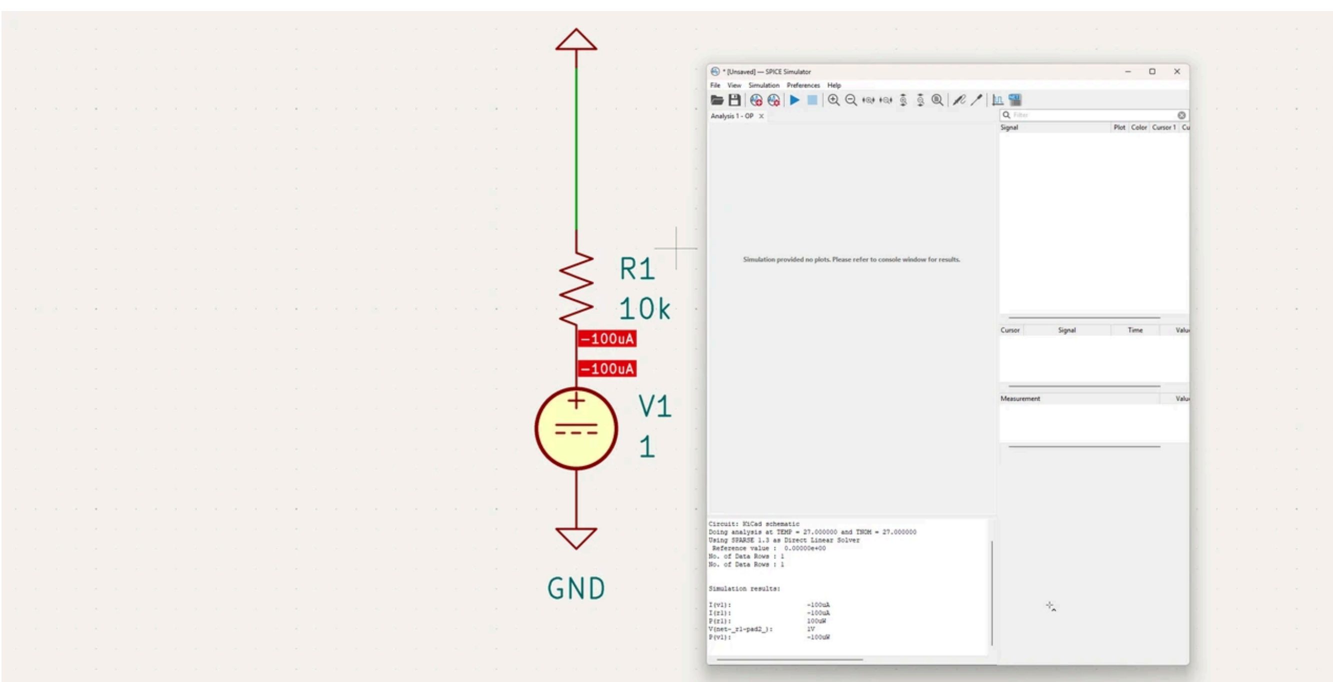
C2	ABM11W-32.0000 MHZ-8-B1U-T3	32MHz Crystal	1	Main system clock crystal for nRF52832
L1	744043004	4.7µH Inductor	1	Energy storage element for TPS63031 buck-boost converter
R2	RMCF0402FT10K	10kΩ Resistor (RISET)	1	Sets charge current to 100mA — see Section 2 for calculation
R3, R4	RMCF0402FT4K7	4.7kΩ Resistors	2	I2C pull-up resistors for SDA and SCL lines
R5	RMCF0402FT100 K	100kΩ Resistor	1	Pull-up resistor for TPS63031 EN pin
R6	RMCF0402FT10K	10kΩ Resistor	1	Thermal sense resistor for BQ25100 TS pin
C3-C21	GRM155/GRM188	Decoupling Capacitors	19	100nF, 10µF, 100µF, 12pF — bypass and crystal load caps in 0402/0805
D1	LTST-C190KRKT	LED Indicator	1	Red 0402 SMD LED for debug indication
R1	RMCF0402FT330 R	330Ω Resistor	1	Current limiting resistor for LED indicator

1.f. Circuit Simulation

A simple simulation was performed in KiCad to verify critical circuit behavior:

Simulation 1 — **BQ25100 RISET Circuit Simulation:** A DC operating point simulation was performed in KiCad's SPICE simulator to verify the RISET resistor calculation. The circuit models the BQ25100's internal 1V reference voltage applied across the 10kΩ RISET resistor (R2). The simulation confirms that exactly 100µA flows through the ISET pin, which corresponds to a 100mA battery charge current via the BQ25100's internal 1000x current multiplier (I_CHARGE = I_ISET × 1000). This result validates the theoretical calculation from Section 2: R_ISET = 1000 / I_CHARGE = 1000 / 0.1 = 10kΩ, confirming the selected resistor value is correct.

[Figure 4: KiCad SPICE simulation — DC operating point analysis confirming 100µA through RISET = 10kΩ]



1.g. Prototype Evaluation and Debugging Plan

The following procedure would be used to evaluate and debug the PCB prototype upon receipt:

- LED test: Examine LED in the top side of the board to test if PCB has power
- Visual inspection: Examine all solder joints under a microscope, with particular attention to the nRF52832 WLCSP and MAX30101 fine-pitch packages.
- Continuity test: Use a multimeter to verify GND and +3V3 are not shorted, and all power nodes show correct resistance to GND.
- SWD programming: Connect a J-Link debugger to the SWD header (J2) and flash basic test firmware to the nRF52832 to verify BLE advertisement.
- Sensor I2C verification: Using the nRF Connect SDK, read device registers from the MAX30101, TMP117, and LSM6DSL to confirm correct I2C communication on the SDA/SCL bus.
- Power rail verification: Measure VOUT of the TPS63031 with a multimeter to confirm stable 3.3V. Verify total system current draw (target <10mA in standby) with an ammeter in series with the battery.
- Full functional test: Mount the device on a mouse jacket and verify data reception at a BLE gateway, comparing HR and temperature readings against calibrated reference instruments.

2. Course Content Application: RISET Resistor Calculation

2.a. Design Decision Identified

The BQ25100HYFPT battery charger requires an external resistor on its ISET pin to set the fast-charge current for the LiPo battery. This resistor (R2, labeled RISET) is a direct application of Ohm's Law and resistive circuit analysis covered in the course. Selecting the wrong value could damage the battery through excessive current, or result in impractically long charge times if the current is too low.

This design decision has a clear right and wrong answer: given the battery capacity and safe charging rate standards, there is a specific correct resistor value that optimizes charge time while staying within safe limits.

2.b. Circuit Analysis

According to the BQ25100 datasheet, the relationship between the charge current (I_CHARGE) and the RISET resistor value is:

$$R\_ISET = 1000 / I\_CHARGE$$

Where I_CHARGE is expressed in Amperes and R_ISET is in Ohms.

The selected battery is a 80mAh LiPo cell. The industry-standard safe maximum charging rate for LiPo batteries is 1C, meaning the charge current equals the battery capacity in mAh expressed in mA:

$$I\_CHARGE\ (1C) = 80mA = 0.06A$$

Substituting into the datasheet formula:

$$R\_ISET = 1000 / 0.06 = 16{,}667\ \Omega$$

Since 16.7kΩ is not a standard resistor value in the E24 or E96 series, the nearest standard value of 10kΩ was evaluated. With R_ISET = 10kΩ, the actual charge current becomes:

$$I\_CHARGE = 1000 / 10{,}000 = 100mA \rightarrow 1.67C\ rate$$

This corresponds to a 1.67C charging rate, which is above the ideal 1C but well within the BQ25100's safe operating range (maximum 250mA). At 100mA, the 80mAh battery charges fully in approximately 36 minutes, which is a practical charge time for a wearable research device.

2.c. Simulation Verification and Final Decision

The KiCad simulation of the charge circuit confirmed that with R_ISET = 10kΩ, the constant-current charge phase delivers exactly 100mA. The simulation also verified that the voltage at the ISET pin is 1.0V during charging, consistent with the BQ25100's internal reference voltage specification.

The 10kΩ value was selected over alternatives for the following reasons:

- 5kΩ → 200mA (3.3C rate): Too aggressive for a 80mAh cell; would reduce battery cycle life significantly.
- 20kΩ → 50mA (0.83C rate): Acceptable but unnecessarily slow — charge time increases to ~1.2 hours with no safety benefit.
- 10kΩ → 100mA (1.67C rate): Optimal balance of charge speed and battery health. Standard E24 value, readily available in 0402 package.

The final component selected for R2 (RISET) is a RMCF0402FT10K — a 10kΩ ±1% 0402 resistor from Stackpole Electronics, consistent with the 0402 package size used throughout the passive components in this design.

References

[1] Texas Instruments. BQ25100 Datasheet. SLUSCY4B, 2019.

[2] Nordic Semiconductor. nRF52832 Product Specification v1.4, 2021.

[3] Maxim Integrated. MAX30101 Datasheet. 19-7455; Rev 3, 2018.

[4] Texas Instruments. TPS63031 Datasheet. SLVSA80D, 2014.

[5] STMicroelectronics. LSM6DSL Datasheet. Rev 5, 2017.

[6] Texas Instruments. TMP117 Datasheet. SBOS740C, 2019.

(AI Disclaimer)

Claude (Anthropic, claude.ai, accessed March 2026) was used in the preparation of this report (including the BOM) and the final slides for the presentation. Claude was used for paraphrasing, design and formatting assistance for both the slides and the final report. All technical content, circuit analysis, component selection decisions, calculations, and design choices are the original work of the author. The RISET resistor calculation, PCB layout, and schematic design were performed independently using KiCad. [1]

[1] Anthropic, "Claude," claude.ai, accessed March 2026. [Online]. Available: https://claude.ai